YBS417 · NETWORK MANAGEMENT AND SECURITY

Secure Multi-Site Enterprise Network for Anatolia Logistics

Design, subnetting, configuration, security and traffic analysis in Cisco Packet Tracer 9.0

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192.168.10.0/24

01 · THE SCENARIO

One company, two sites, one /24 block

Anatolia Logistics has a Headquarters with five departments, a small Branch Office and an Internet uplink. It needs separated departments, automatic addressing, central management and protection from guests and LAN attacks.

Segment

Six VLANs, one per department and site

Size right

VLSM: every subnet fits its real host count

Manage

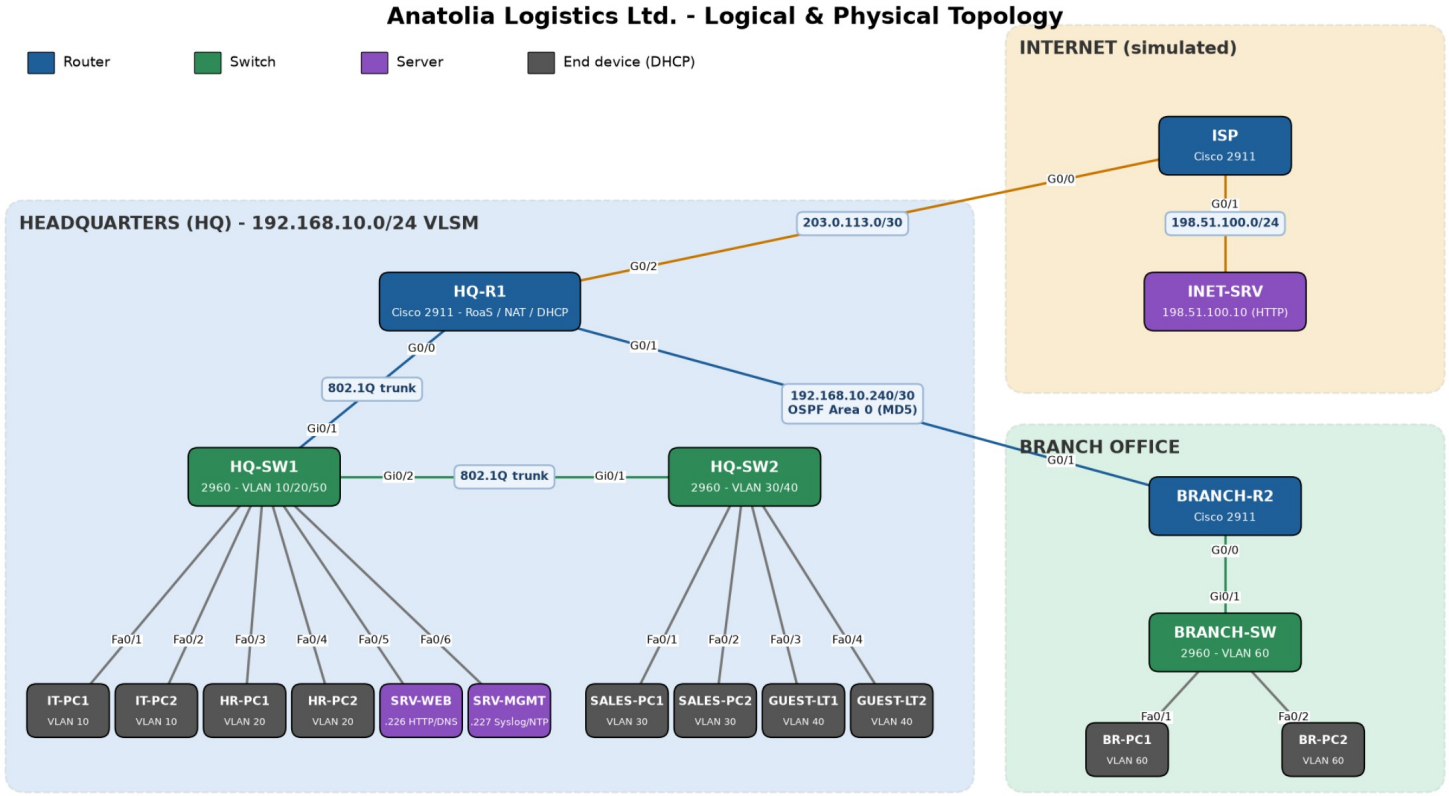
DHCP, Syslog, NTP, TFTP backups, SSH

Secure

Controls at device, port, VLAN, routing and traffic layers

HQ, Branch and a simulated ISP

3 × Cisco 2911 · 3 × Cisco 2960
3 servers · 12 end devices



03 · ADDRESSING PLAN

Seven subnets from 192.168.10.0/24

Segment	Hosts	Network	Gateway	Broadcast
Guest · VLAN 40	50	192.168.10.0/26	.1	.63
Sales · VLAN 30	40	192.168.10.64/26	.65	.127
IT / Mgmt · VLAN 10	25	192.168.10.128/27	.129	.159
HR · VLAN 20	20	192.168.10.160/27	.161	.191
Branch · VLAN 60	18	192.168.10.192/27	.193	.223
Servers · VLAN 50	8	192.168.10.224/28	.225	.239
HQ–Branch WAN	2	192.168.10.240/30	.241 / .242	.243

Largest first, smallest block that fits

Need

50 hosts

Guest VLAN

Host bits

$2^6 - 2 = 62$

$2^5 - 2 = 30$ is too small

Prefix

/26

255.255.255.192

Block of 64

.0 – .63

Next subnet starts at .64

244 / 256

addresses allocated · 95% used

With fixed /27 subnets the 50-host Guest VLAN would not fit, and the WAN link would waste 28 addresses.

04 · CONFIGURATION

Core services that keep it running

Router-on-a-stick

One trunk, five 802.1Q sub-interfaces on HQ-R1

DHCP

Router pools per VLAN, static ranges excluded

OSPF area 0

HQ ↔ Branch, default route sent to Branch

NAT / PAT

All hosts share 203.0.113.2 to the Internet

DNS + Web

intranet.anatolia.local on SRV-WEB

Management

Syslog, NTP and TFTP backups on SRV-MGMT

Five layers of defense, each one stopping a different attack.

Device | Switch port | VLAN / STP | Routing | Traffic (ACL)

Every threat matched to a control

Layer	Threat	Control
Device	Password guessing, sniffing	enable secret, login block-for, SSH v2 only
Device	Unauthorized management	VTY access-class: only VLAN 10 (IT)
Switch port	Rogue device, MAC flooding	Port security: 1 sticky MAC, shutdown
Switch port	Rogue DHCP server	DHCP snooping, only uplinks trusted
VLAN / STP	VLAN hopping, STP attacks	Native VLAN 999, DTP off, BPDU Guard, unused ports shut
Routing	Fake OSPF neighbors	OSPF MD5 authentication, passive LANs
Traffic	Guests and others reaching data	GUEST-IN and HR-PROTECT extended ACLs

Two ACLs, placed for different reasons

GUEST-IN

inbound · G0/0.40

Near the source: guest packets are dropped before they cross the router.

```
permit DHCP requests
permit DNS → SRV-WEB
permit ping → own gateway
deny → 192.168.10.0/24
permit → Internet
```

HR-PROTECT

outbound · G0/0.20

Near the destination: one list guards HR from Sales, Guest and Branch.

```
permit IT VLAN → HR
permit Servers → HR
permit tcp established
permit icmp echo-reply
deny everything else
```

The policy holds: allowed and blocked flows

Source	Test	Result	Why
IT-PC1	ping HR-PC1	Allowed	IT permitted by HR-PROTECT
SALES-PC1	ping HR-PC1	Blocked	HR-PROTECT deny
BR-PC1	intranet.anatolia.local	Allowed	OSPF + DNS + HTTP
GUEST-LT1	www.internet.com	Allowed	DNS permit + PAT
GUEST-LT1	intranet.anatolia.local	Blocked	GUEST-IN deny
IT-PC1	ssh netadmin@HQ-R1	Allowed	VTY ACL permits VLAN 10
SALES-PC1	ssh netadmin@HQ-R1	Blocked	VTY ACL, refused
IT-PC1	telnet HQ-R1	Blocked	SSH-only VTY lines

06 · ATTACK TESTS

We attacked our own network

Rogue device

A new laptop replaced IT-PC2 on Fa0/2.

Result: port err-disabled, PSECURE_VIOLATION logged.

Rogue DHCP

A fake DHCP server in the Guest VLAN offered gateway .50.

Result: offers dropped; clients still got gateway .1.

Brute force

Three wrong SSH passwords within 60 seconds.

Result: router refused all logins for 120 seconds.

What Simulation mode showed us

DHCP	Discover → Offer → Request → Ack, inspected by DHCP snooping
802.1Q	Frame enters the trunk tagged 30, returns tagged 10
NAT	Source 192.168.10.x becomes 203.0.113.2 with a new port
ACL drop	Sales → HR packet dropped on G0/0.20, ICMP unreachable returned
OSPF	Hellos only on the WAN link, carrying MD5 authentication

07 · CHALLENGES

Four problems we had to solve

PCs lost DHCP after snooping

Fix: no ip dhcp snooping information option, trusted uplinks

HR could not browse

Fix: permit established and echo-reply in HR-PROTECT

Guests could not resolve names

Fix: DNS permit placed above the deny line (ACL order)

Replaced PC disabled the port

Fix: documented recovery: clear sticky MAC, shutdown / no shutdown

08 · CONCLUSION

A segmented, managed and layered-secure network on one /24

We delivered

7 VLSM subnets · 6 VLANs · OSPF + PAT

Central DHCP, Syslog, NTP, TFTP

13 security controls, 16 tests, 3 attack tests

Next steps

HSRP and a backup WAN link

Cisco ASA firewall, RADIUS / TACACS+

WPA2-Enterprise Wi-Fi, IPv6 dual stack

Live demo & questions